# TilapiaSupplyChain

*Farm-to-Market Supply Chain Tracking System*

Scenario: Tilapia Farming | Blockchain: Ethereum | Tool: Remix IDE

## 1. Environment Setup

The contract was deployed using Remix IDE connected to a local Geth Testnet (Chain ID 1337) running on 127.0.0.1:8545.

**Output: Geth Testnet Running — Available Accounts & Private Keys**

```
Starting RPC server

Available Accounts
==================
(0) 0x137fC646f464c116adE85f42Efa7648b2f9b1b0a (1000 ETH)
(1) 0x4D9313A492cE20f15f0cF6c123B291fA635D2DE5 (1000 ETH)
(2) 0x520cb4aFADE8a19e021A451406b95Fc6843095b2 (1000 ETH)
(3) 0xa7E35ACE9534eb1549cbAe2Ae2F1A4f34c0Ab29B (1000 ETH)
(4) 0x92702263A3F2F7553B2FBc792795e06bA12CEC72 (1000 ETH)
(5) 0xee35D5211D1bc9eb12B30bd7548B084b2d5FBC1E (1000 ETH)
(6) 0xeCDC1C12a1e74De7e96Fa0A2c671aDbF21ff872B (1000 ETH)
(7) 0x18e167486EDD0d174D3b4A561d40Ba96fae77872 (1000 ETH)
(8) 0xA7c754715FF7713C0B675893FDB282acd8B1bAf4 (1000 ETH)
(9) 0x59FD77c4dd246e497155A0Bc34bdAc1e53De2Da3 (1000 ETH)

Private Keys
==================
(0) 0xfb3e958af642ecee4ecb0002e8eb3fd1722b40d38990cfff34d5074c22a21063
(1) 0xd734453f6dfa3a5b7353604d299dcbe0bd888d9bfe472ab53c55f789f887bd23
(2) 0x626b07d5ca48e63c82e376ab3bfe1cffbdaa59ad6228109745bfdde619c46c2f
(3) 0xe71518782a6ac7e673bc3508b6594cdd86387bdf8e43970780ff852279ee3ab4
(4) 0x2b45e6c74b88da0f71ce1903d20d36fafb2e9da44edcc2653625b98c819323ca
(5) 0xd16ad1ca582ba49773ae3fafcbef8830dc99339b2afd704197ca606f10753da8
(6) 0x5e97dd63c341c214f84d8c9a7a5f98f074950b2e3b3c3899eedc60020fc98730
(7) 0xa10a51a0673a8fc92865d16784577e6a12c67680ec4b8699c346c851e97d48cc
(8) 0xe739f14b8f597bea63839a0582a2de6d1cc5a940c2715d098b3488af596f1fe4
(9) 0x54cb997d221659abf0585b8ac4da7eee9ced9beeb39e0c2822739852fe9bacbc
```

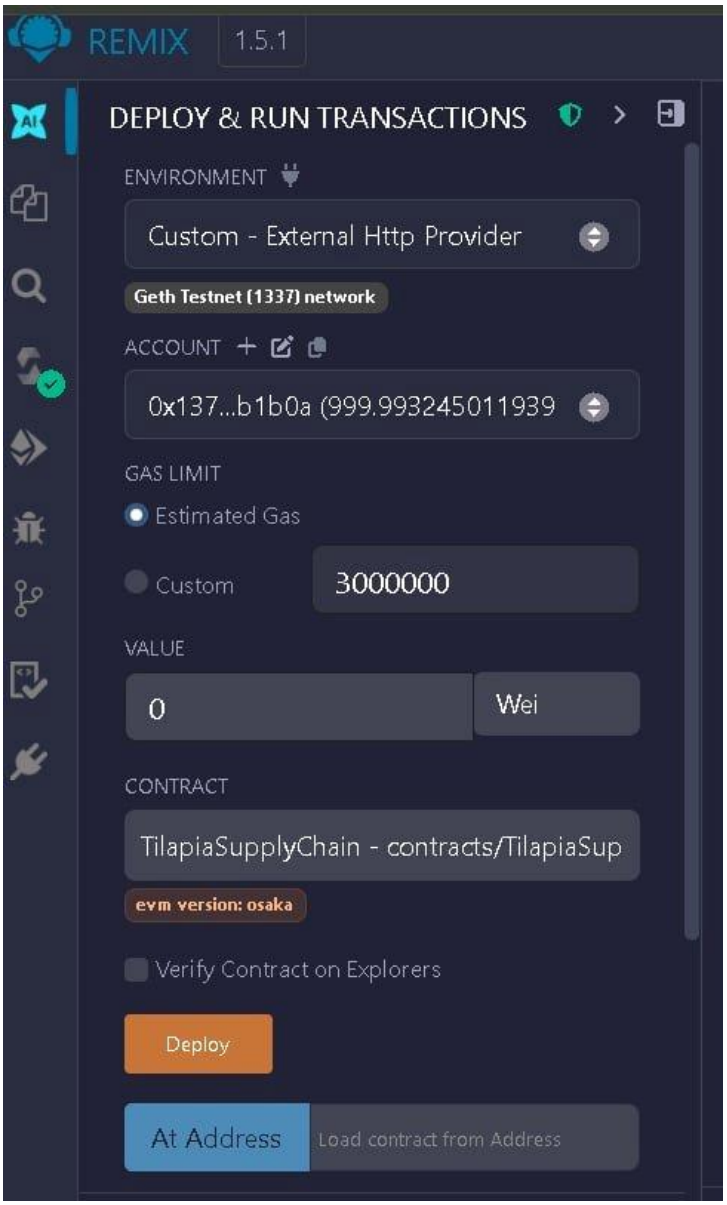
**Output: RPC Server Active — Listening on 127.0.0.1:8545**

```
HD Wallet
================
Mnemonic:      river acquire laptop such nature earth combine habit antenna wise c
Base HD Path:  m/44'/60'/0'/0/{account_index}

Default Gas Price
================
2000000000

BlockGas Limit
================
30000000

Call Gas Limit
================
50000000

Chain
================
Hardfork: shanghai
Id:       1337

RPC Listening on 127.0.0.1:8545
eth_chainId
eth_chainId
eth_getBlockByNumber
eth_getBlockByNumber
eth_chainId
eth_accounts
```

## 2. Remix IDE Configuration

Remix was set to use Custom - External HTTP Provider to connect to the local Geth node. The deployer account (0x137...b1b0a) was selected with Estimated Gas enabled.

**Output: Remix Deploy & Run Transactions Panel**




## 3. Contract Deployment

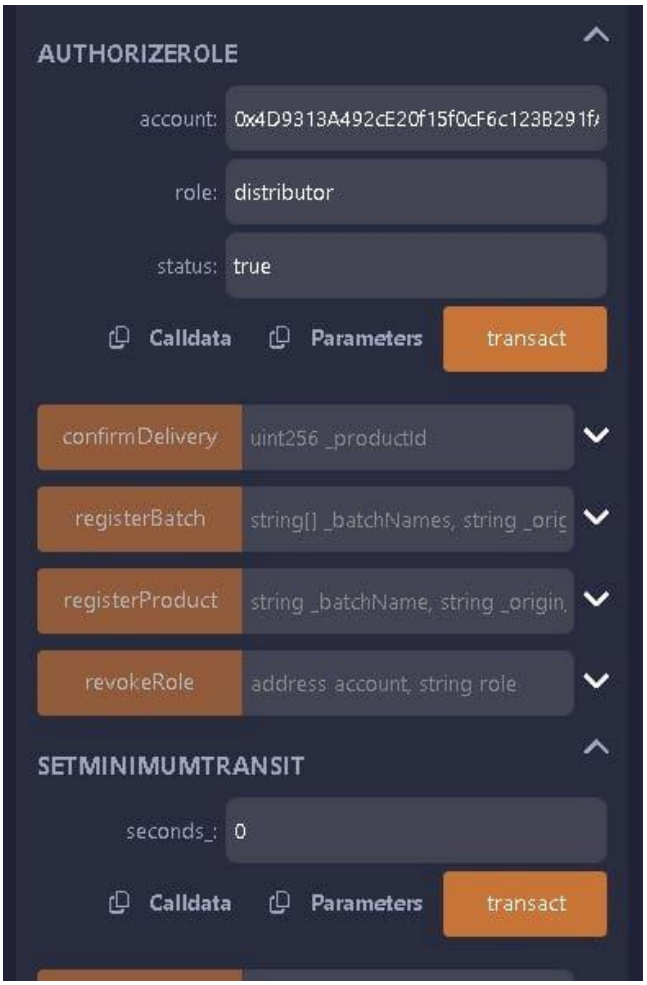
The TilapiaSupplyChain contract was deployed successfully. The transaction was mined at block number 2 with a cost of 3,207,472 gas.

**Output: Deployment Transaction Result**

## 4. Contract Functions & Role Authorization

After deployment, the contract exposes the following functions: confirmDelivery, registerBatch, registerProduct, and revokeRole. The AUTHORIZEROLE panel was used to grant the distributor role to account 0x4D9313A4...

**Output: Contract Functions Panel & AUTHORIZEROLE (role: distributor, status: true)**




## 5. How the Supply Chain Works

The contract tracks tilapia products through four stages:

1. Farmer registers a product or batch (status: Created)
2. Farmer transfers to Distributor (status: In Transit)
3. Distributor transfers to Retailer (status: Delivered)
4. Retailer marks product as sold (status: Sold)

Only authorized roles can register or transfer products. The admin uses authorizeRole() to grant access and revokeRole() to remove it. All ownership history is stored on-chain and can be queried by anyone.